Eating Cherries

Input file:	`standard input`
Output file:	`standard output`
Time limit:	2 seconds
Memory limit:	1024 megabytes

Jiangqiao loves eating cherries! Before the cherries ripened, he had already arrived at the cherry tree...

A cherry tree consists of n nodes (numbered 1 to n) and $n-1$ branches, which connect all the nodes together.

Jiangqiao classifies the cherries on the whole cherry tree into m types (numbered 1 to m) based on size, color, sweetness, etc., where node i has k_i types of cherries.

A **main branch** is the path from the root node to a certain leaf node without repeating any node.

That is, for root node u, the main branch from u to a leaf node v is a node sequence $[u = x_0, x_1, x_2, \ldots, x_s = v]$, where node x_i is connected to node x_{i+1} $(i < s)$ by a branch, and $x_i \neq x_j$ whenever $i \neq j$.

A **leaf node** refers to a node that is connected to only one branch with other nodes. Specifically, **the root node is not a leaf node**.

Obviously, for each leaf node, the main branch from the root to it is uniquely determined.

Jiangqiao can cast magic. **Each spell** can choose a leaf node and **arbitrarily change** the cherry types and the number of types on that node (including making it cherry-free).

He wants to know, for each node $1, 2, ..., n$ serving as the root, the **minimum number of spells** needed so that the **set of cherry types** on every main branch is the same.

Input

The first line contains two integers n and m ($2 \leq n \leq 2 \times 10^5$, $1 \leq m \leq 6$) — the number of nodes and the number of cherry types.

Each of the next $n-1$ lines contains two integers u_i and v_i ($1 \leq u_i, v_i \leq n$, $u_i \neq v_i$) — a branch on the tree.

Each of the next n lines describes a node. The i-th of these lines starts with a non-negative integer k_i ($0 \leq k_i \leq m$) — the number of cherry types at node i. If $k_i > 0$, it is followed by k_i distinct integers $t_{i,1}, t_{i,2}, \ldots, t_{i,k_i}$ ($1 \leq t_{i,j} \leq m$) — the cherry types.

It is guaranteed that all nodes are connected by the branches.

Output

A single line containing n non-negative integers, where the i-th integer is the answer when node i is the root.

Example

standard input	standard output
5 2 1 2 2 3 2 4 4 5 1 1 1 1 2 1 2 0 1 1	1 1 0 1 1